Supercooled Water Cloud Identification From Geostationary Satellite Thermal Infrared Observations and Its Application in the Sun Glint Region

Haoyang Fu, Feng Zhang, Bin Guo, and Wenwen Li

***Abstract*— Supercooled water clouds (SWCs) are prevalent in the atmosphere and crucial for global and local radiation balance, aviation safety, and weather modification techniques like artificial precipitation. Therefore, there is an imperative need for the continuous and precise monitoring of SWCs at high temporal and spatial resolution, encompassing observations under all sky conditions. This study aims to enhance the identification of SWCs by leveraging thermal infrared (TIR) channels of the Himawari-8 geostationary satellite, regardless of solar illumination or sun glint effects, which can be problematic for reflectivity bands. Principal component analysis (PCA) is utilized to perform a sensitivity analysis on a dataset comprising TIR bands from the Himawari-8 satellite and labels derived from the Cloud-Aerosol LiDAR and Infrared Pathfinder Satellite Observation (CALIPSO) cloud profile products, to assess the efficacy of the data in differentiating between supercooled water and ice clouds (ICs). Subsequently, machine learning techniques are employed to develop an all-day SWC identification model. The model is assessed using a time-independent dataset, yielding an overall accuracy rate for cloud phase (CPH) identification of over 90%, as well as high performance for detecting SWCs. The model demonstrates consistent performance across various surfaces, times of day, and seasons. Notably, it outperforms traditional algorithms that rely on reflectivity bands by accurately identifying SWCs even in sun glint regions, thus improving the reliability of CPH detection for applications in meteorology, climate research, and aviation safety.**

***Index Terms*— Cloud phase (CPH), Himawari-8, supercooled water clouds (SWCs), thermal infrared (TIR).**

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I. Introduction

CLOUDS envelop nearly 70% of the Earth's surface and profoundly influence the global and local energy balance, as well as water cycles [1], [2], [3], [4]. Precise characterization of clouds is vital for weather and climate models, which are essential for predicting precipitation, calculating surface radiation, forecasting extreme weather, and understanding climate change [5], [6], [7], [8], [9]. Clouds are classified into ice clouds (ICs) and water clouds based on the thermodynamic phase of their particles, that is a critical factor in the accurate retrieval of multiple cloud properties [10], [11]. Liquid water clouds below 0 °C without nucleating into ice are known as supercooled water clouds (SWCs) [12], [13]. The SWCs are ubiquitous in the atmosphere and play a crucial role in the local and global radiation balance, especially in weather and climate models [14], [15]. Moreover, SWCs can readily freeze upon contact with aircraft surfaces and engines, posing significant risks to flight safety [16], [17]. Additionally, these clouds are capable of producing precipitation through seeding, making their accurate identification crucial for enhancing artificial precipitation techniques [18], [19].

Satellite remote sensing technology has undergone rapid advancement in recent years, leading to its prevalent use for extensive, both local and global, cloud detection and physical property retrievals via diverse satellite platforms and sensing techniques [20], [21], [22]. High-performance sensors on polar-orbiting platforms at low altitudes, such as the Moderate Resolution Imaging Spectroradiometer (MODIS) and the Visible Infrared Imaging Radiometer Suite (VIIRS), can provide multiband, high-resolution observations with a high signal-to-noise ratio, enabling daily global coverage. New advanced sensors equipped on geostationary satellites like the Advanced Himawari Imager (AHI) on the Himawari-8/9, the Advanced Baseline Imager (ABI) on the Geostationary Operational Environmental Satellite—R Series (GOES-R), and Advanced Geostationary Radiation Imager (AGRI) on FengYun-4 satellites offer higher spatial resolution and accuracy compared to their predecessors [23]. Unlike polar-orbiting satellites that pass over at a consistent local time, geostationary satellites excel in providing continuous, high-temporal resolution observations over large areas [24]. Given its versatility and effectiveness in atmospheric research, satellite remote sensing has also been employed to study SWCs, yielding

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notable achievements. Pavolonis [25] crafted an algorithm that discerns cloud phase (CPH) including SWC after cloud detection for the ABI by integrating cloud temperature and cloud thickness. The AHI equipped on the Himawari-8/9 features 16 channels covering a spectrum from visible and near-infrared (VNIR) to thermal infrared (TIR) with spatial resolutions ranging from 0.5 to 2 km. It enables continuous monitoring of clouds and atmospheric conditions at 10-min intervals [26], [27], [28]. While official CPH products, including ice, water, and mixed phases, are available, they do not specifically identify SWCs. Wang et al. [29] leveraged Himawari-8's Level 2 cloud products, such as CPH, cloud optical thickness (COT), cloud effective radius (CER), and cloud top temperature (CTT), employing threshold settings to detect SWCs. This method achieved a high detection rate but also resulted in a significant false alarm rate, due in part to limitations inherent in the initial cloud products. Zhou et al. [30] introduced a SWC identification method based on the VIIRS by exploiting the reflectance difference between two shortwave infrared channels at 1.61 and 2.25 μm, as well as the brightness temperature difference between two longwave infrared channels at 8.5 and 11 μm. However, it is restricted by the polar orbit, preventing continuous, high-frequency monitoring of the same region. The optical properties of ice and liquid clouds are distinctly different in some specific short-wavelength reflectance bands, allowing for straightforward differentiation [31], [32], [33]. However, these existing algorithm's effectiveness is compromised at night and in regions affected by sun glint [34], [35], [36], [37].

LiDAR, unlike passive sensors that rely on sunlight reflection or blackbody radiation for observation, utilizes active laser signals to detect clouds with superior accuracy, unaffected by solar illumination [38], [39], [40]. Employing the Cloud-Aerosol LiDAR with Orthogonal Polarization (CALIOP) and Imaging Infrared Radiometer (IIR) sensors on the Cloud-Aerosol LiDAR and Infrared Pathfinder Satellite Observation (CALIPSO) satellite, in conjunction with MODIS also from the Afternoon satellite constellation (A-Train), Hogan et al. [41] and Hu et al. [42], [43] have enhanced the identification and analysis of SWCs, shedding light on their spatial distribution patterns to some extent. Nevertheless, these results face limitations due to LiDAR's constrained spatiotemporal observation capabilities. The TIR channels on passive sensors detect objects' blackbody radiation, enabling observation without sunlight and facilitating all-sky cloud retrieval [44], [45], [46]. However, due to the weak signal of the radiation and the complex nonlinear relationships with cloud physical processes, just using simple thresholds for cloud detection and phase identification in various conditions is challenging. Advanced artificial intelligence technologies have made significant strides in recent years, offering robust tools for developing remote sensing algorithms. Machine learning methods can leverage multichannel data to extract more information, enhancing algorithm performance over traditional threshold-based approaches [47], [48], [49]. Accurate cloud measurements from active remote sensing sensors, like LiDAR, provide valuable labels and benchmarks for training and testing machine learning models in cloud retrieval using passive observations [50], [51], [52], thereby increasing accuracy and extending the application scope of passive remote sensing [53]. Li et al. [54] utilized neural networks for cloud detection and phase classification with the AHI sensor's TIR channels on the Himawari-8 satellite. They incorporated clear sky (CS) brightness temperature data from radiation transfer simulations to identify all-sky CPH states and multilayer clouds using the infrared brightness temperature channels, although their study did not cover SWCs. Therefore, current applications of passive remote sensing in geostationary orbit reveal the potential for further research into SWCs, particularly in developing precise all-sky products.

In this study, the TIR bands from the Himawari-8's AHI with central wavelengths ranging from 5 to 14 μm were leveraged to identify SWCs under all-day conditions. Utilizing the CALIPSO Level 2 Aerosol/Cloud Profile products, labels with four categories were constructed as benchmarks for both training and evaluating the machine learning model. Section II details the data utilized and the methodology for dataset construction. Section III presents a sensitivity analysis using principal component analysis (PCA) to assess the input features' sensitivity to SWCs and ICs. Section IV describes the machine learning model and evaluation method employed by the algorithm. Section V discusses the model's overall results, applications to sun glint regions, and considerations under various conditions. Finally, Section VI provides a comprehensive summary of this article.

II. Data

A. Data Source

1) TIR Observations From Himawari-8/AHI: The Himawari-8/AHI is equipped with 16 spectral bands, encompassing six VNIR and ten TIR channels, offering a spatial resolution between 0.5 and 2 km, and allows for continuous observations at 10-min intervals. For the purpose of this study, only bands 8–16 with central wavelengths at 6.2, 6.9, 7.3, 8.6, 9.6, 10.4, 11.2, 12.4, and 13.3 μm in Himawari-8/AHI standard data were utilized. Consequently, band 7 at 3.9 μm susceptible to sunlight was deliberately excluded. Moreover, the satellite zenith angle (SAZ) was employed to capture the influence of different path lengths on radiative transfer processes.

2) CALIPSO Level 2 Aerosol/Cloud Profile Products: The CALIPSO Level 2 Aerosol/Cloud Profile Data employs dual-polarization channels at 532 nm and a single channel at 1064 nm from CALIOP, providing detailed CPH profile information. It achieves a horizontal resolution of 5 km and a vertical resolution of 60 m for altitudes between 0 and 20.1 km, and a vertical resolution of 180 m for altitudes between 20.2 and 30.1 km [42]. Additionally, temperature profiles derived from the Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) also contained in the products, were utilized to identify SWCs in this study.

3) Auxiliary Data: The European Centre for Medium-Range Weather Forecasts Reanalysis version 5 (ERA-5) supplies hourly estimates on atmospheric, land, and oceanic climate variables across single and pressure levels at a spatial resolution of 0.25° [55]. Selected single-level data included surface pressure, skin temperature, and total column water

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vapor, while the chosen pressure level data encompassed ozone mass mixing ratios, specific humidity, and temperature at altitudes of 1000, 850, 500, and 200 hPa. Additionally, surface emissivity data is derived from the Terra MODIS Land Surface Temperature/Emissivity 8-Day and Monthly data Version 6 (MOD11C2 and MOD11C3), covering MODIS bands 20, 22, 23, 29, 31, and 32 with a spatial resolution of 0.05°. These elements, which are crucial to upward TIR radiation, represent the influence of surface and atmospheric conditions on the brightness temperature observed. That would help refine the model's constraints to enhance its accuracy.

B. Dataset Construction

The process of constructing the dataset was bifurcated into two primary steps: label generation and data matching, with the generation of high-precision labels being pivotal for the efficacy of machine learning algorithms. This study leveraged CPH profile information from the CALIPSO Level 2 Aerosol/Cloud Profile Data, augmented with temperature profile data, to classify liquid clouds below 0 °C as SWCs. Subsequently, profiles characterized by a singular CPH type were preserved and tagged according to their specific CPH, thereby reducing the dimensions. The resulting classification framework comprises four categories: CS, warm water cloud (WWC), SWC, and IC. While LiDAR exhibits excellent detection capabilities through thin clouds, it faces substantial attenuation when encountering thick clouds. Consequently, CALIPSO/CALIOP can accurately reflect cloud top information for both thin and thick clouds, closely matching the penetration capabilities in passive TIR channel observations.

In this research, the spatiotemporal information of CALIPSO was utilized as standards to align with others including Himawari-8/AHI data, aiming to minimize discrepancies attributable to variations in satellite orbits, observation techniques, data collection frequencies, and observation timings. For spatial matching, CALIPSO/CALIOP observations are nearly perpendicular to the Earth's surface, with a SAZ of approximately 0.3°. In contrast, the SAZ for Himawari-8/AHI increases with distance from the nadir point, leading to potential misalignments. Although previous studies have addressed geometric parallax correction using cloud top height, challenges persist in adjusting for significant SAZ differences and atmospheric conditions across various altitudes. To address these challenges, this study intensified the quality control of CALIPSO-derived labels by preserving only intermediate samples where at least five consecutive pixels of the same type were detected along an observational track. This rigorous quality control approach successfully excluded samples with substantial uncertainty, ensuring the integrity and accuracy of the classified data. Furthermore, by capitalizing on the consistent physical properties of clouds in proximate clusters, discrepancies caused by diverse observation methods, angles, and spatial resolutions among multisource data were effectively reduced. Consequently, this strategy not only ensured the accuracy of our research but also obviated the computational demands associated with geometric parallax corrections. In the full disk observations conducted by Himawari-8/AHI, the imaging time varies for each pixel. Specifically, the time discrepancy can reach up to nearly 9 min between pixels at latitudes from north to south. However, the observation time of each pixel remains relatively constant to others in the same data. To achieve optimal temporal matching, it is essential to take the spatial position of each pixel into account, ensuring minimal differences in the comparative analysis. Following the outlined spatiotemporal matching process, we ensure that the time discrepancy between CALIPSO labels and Himawari-8/AHI observational data is kept under 5 min.

The data employed encompasses the entire year of 2016 and specific months of 2017—January, April, August, and November, as detailed in Table I. All input features were selected for their relevance to upward radiation transfer and the energy received by satellite sensors and were standardized by mean removal and variance scaling. Despite the inherent challenges of merging data from multiple sources, the computational burden and potential inaccuracies typically associated with the radiative transfer model were circumvented.

III. Sensitivity Analysis

In this study, the identification of SWCs primarily utilized the TIR channels of Himawari-8/AHI. The initial steps involved assessing the sensitivity of input variables to SWCs. The TIR channels, influenced by blackbody radiation, are particularly responsive to temperature variations. That allows for the clear distinction between WWCs, which are notably warmer, and other cloud types. However, distinguishing between ICs and SWCs presents challenges due to their overlapping temperature ranges from 0 to −40 °C. Their optical characteristics within the TIR bands of Himawari-8/AHI are significantly influenced by the complex refractive index. Under ideal conditions, the complex refractive indices of SWCs and ICs differ significantly across TIR bands, and the real and imaginary parts are indicative of the medium's radiation scattering and absorption properties, respectively. While the complex refractive index provides insight into the extinction differences between SWCs and ICs in TIR channels, the reality encompasses a broader array of atmospheric, cloud, and surface factors, accompanied by more intricate physical processes. Consequently, this study employed PCA on the dataset constructed to enhance our understanding of the discriminative sensitivity between SWCs and ICs. PCA utilizes orthogonal linear transformation to convert variables of a dataset into linearly independent principal components (PCs), ordered by their variance. This method effectively reduces data dimensionality with minimal information loss by prioritizing components that contain the most information. In this study, the first three PCs account for over 70% of the total variance, indicating they capture most of the dataset's information. Conversely, the fourth component contributes less than 5% to the total variance, affirming the efficiency of PCA in data simplification. Fig. 1 illustrates the distribution of PC1 and PC2 within a narrow range of PC3 values, plotted on a 2-D axis. This visualization clearly differentiates most SWCs from ICs, demonstrating the effectiveness of combining TIR channels with auxiliary data for SWC identification. However, the varying positions of these two types within the narrow

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TABLE I
DATASET COMPOSITION AND FEATURE INFORMATION

	Data source	Features	Spatial resolution	Time resolution
Input	Himawari-8 AHI Standard Data	Bands 8-16 SAZ	2 km	10 min
	ERA-5 reanalysis	**Single levels:** Surface pressure Skin temperature Total column water vapor **Pressure levels:** Ozone mass mixing ratio Specific humidity Temperature	0.25 °	1 hour
	MOD11C2/C3	Emissivities of bands 20, 22, 23, 29, 32 and 33	1 km	8 days/ one month
Label	CALIPSO Level 2 Cloud Profile Data	Cloud Phase	5 km	——

PC3 interval suggest their distributions are not static, making it challenging to achieve accurate separation in the feature space using simple linear hyperplanes.

IV. METHODOLOGY

A. Model Constructions

Sensitivity analysis reveals that SWC classification exhibits significant nonlinear properties. To address this, we employed the extreme gradient boosting (XGBoost) algorithm, a powerful ensemble learning method, for its superior accuracy and robustness, particularly with tabular datasets [56], [57]. The classification model was trained using the entire 2016 dataset, comprising 1 854 704 samples, to mitigate seasonal bias. Given the prevalence of certain cloud types (CS and IC) over others (WWC and SWC), the Synthetic Minority Oversampling Technique (SMOTE) [58] was applied to balance the training dataset. Fig. S1 in the Supplementary Material, shows the training dataset's sample distribution before and after balancing. This approach maintained both diversity and distribution of samples, ensuring convergence and generalization of the model. Fivefold cross validation (fivefold CV) was implemented for optimal hyperparameter tuning with parameters detailed in Table S1 in the Supplementary Material. The flowchart of the SWC detection algorithm using AHI, MODIS, and ERA5 data is shown in Fig. 2.

B. Evaluation Metrics

In this article, general classification evaluation metrics, including accuracy, precision, recall, and F1-score, were used to evaluate the model while drawing a confusion matrix of classification results. The detailed formulas are as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

where true positive (TP) represents the count of correctly identified positive cases. False negative (FN) refers to the count of positive cases incorrectly labeled as negative. False positive (FP) is the number of negative cases mistakenly identified as positive. Lastly, true negative (TN) denotes the count of negative cases accurately classified as negative.

C. Sun Glint Region

Sunlight can be specularly reflected off surfaces with high reflectance, such as water, snow, and clouds, creating a sun glint area in satellite observations under specific angular relationships. This sun glint effect can significantly increase the intensity of solar radiation detected by satellite sensors, potentially leading to saturation and impairing the performance of algorithms or models that rely on VNIR bands. The severity of sun glint is determined by the glint angle, with smaller angles resulting in stronger glint effects. A 0° angle indicates direct specular reflection, and sun glint regions are typically identified when the angle is below a threshold, usually between 25° and 40° [60]. The sun glint angle is calculated using the following formula:

$$\Theta_{\text{glint}} = \cos^{-1}(\cos\theta_S \cos\theta_V - \sin\theta_S \sin\theta_V \cos\phi) \tag{5}$$

where θ_S is the solar zenith angle, θ_V is the SAZ and ϕ represents the relative azimuth angle between the Sun and satellite.

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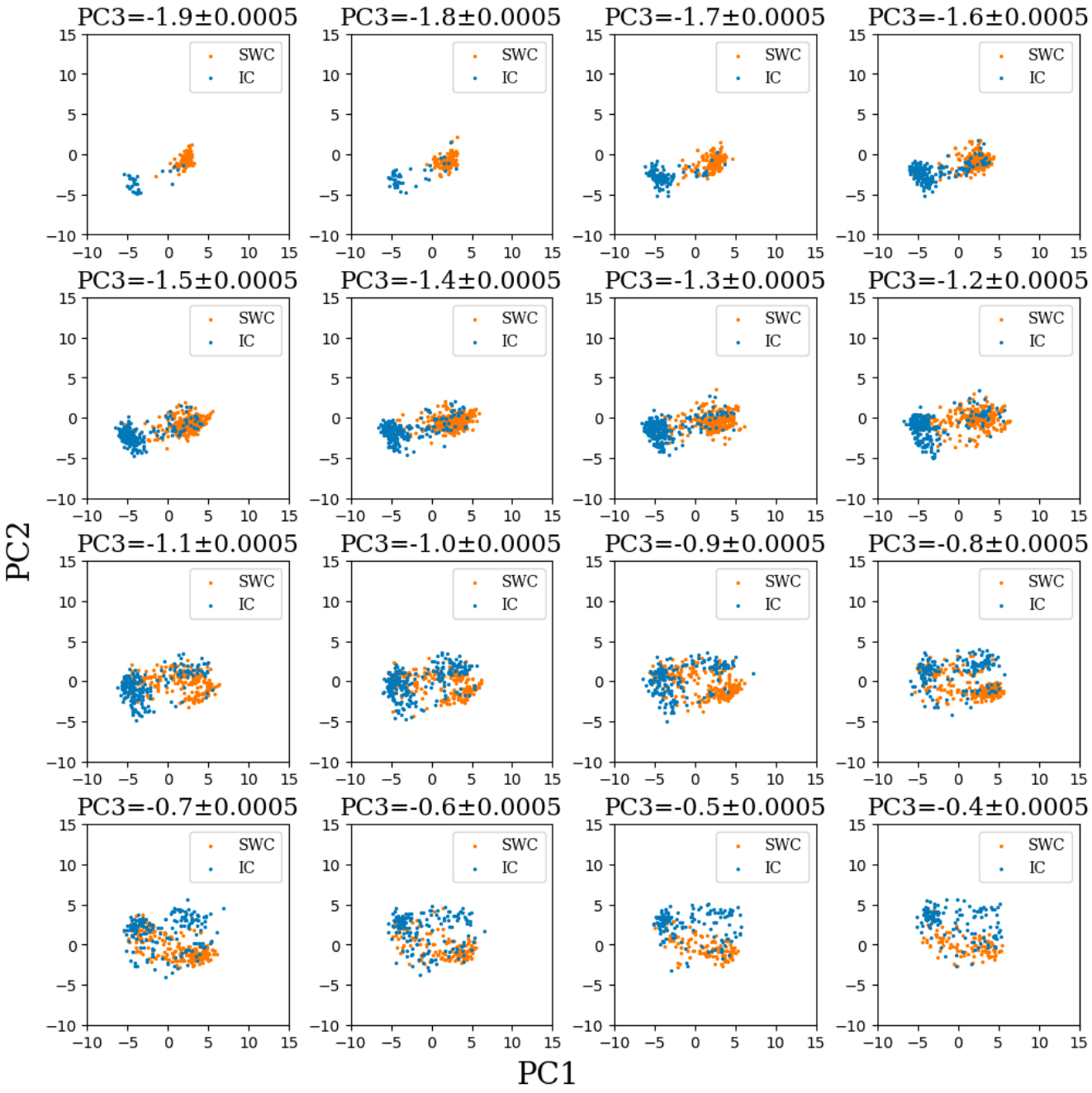


Fig. 1. Distribution of PC1 and PC2 within a narrow range of PC3 values after PCA of the constructed Himawari-8/AHI dataset.

V. Results and Discussion

A. Comprehensive Evaluation of Model and Its Performance in the Sun Glint Region

The model was trained on the comprehensive 2016 dataset, with Himawari-8/AHI data being crucial, especially bands 14, 15, and 13. In comparison, the vapor absorption band 8 is less critical. Auxiliary data such as meteorological fields, surface emissivity, and SAZ have minimal impact on the model. The significance of each input feature, determined by the frequency with which it is used to split the data across all trees, is detailed in Fig. S2 in the Supplementary Material. The trained XGBoost model in this study was evaluated against an independent dataset containing 1 255 656 samples from January, April, August, and November 2017, with CALIPSO labels serving as the benchmark, and the sample distribution of the testing dataset is shown in Fig. S1. The evaluation, illustrated in Fig. 3 via a confusion matrix, demonstrates the model's overall effectiveness. Specifically, it achieves both recall and accuracy rates above 90% in identifying SWCs. The model also effectively distinguishes between other types, though it misclassifies some ICs as CSs. This issue stems from the limited sensitivity of passive TIR channels to ICs with low COT, a challenge not faced by CALIOP due to its higher sensitivity to thin clouds and aerosols. Nonetheless, the

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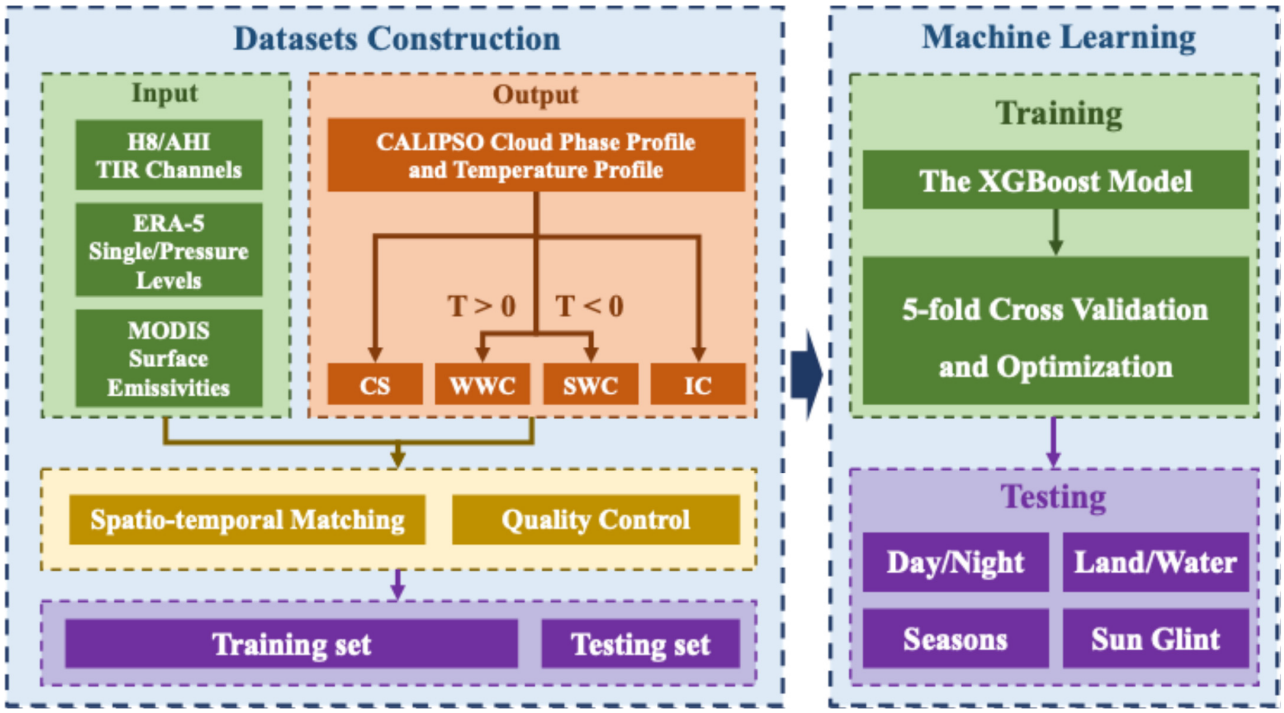
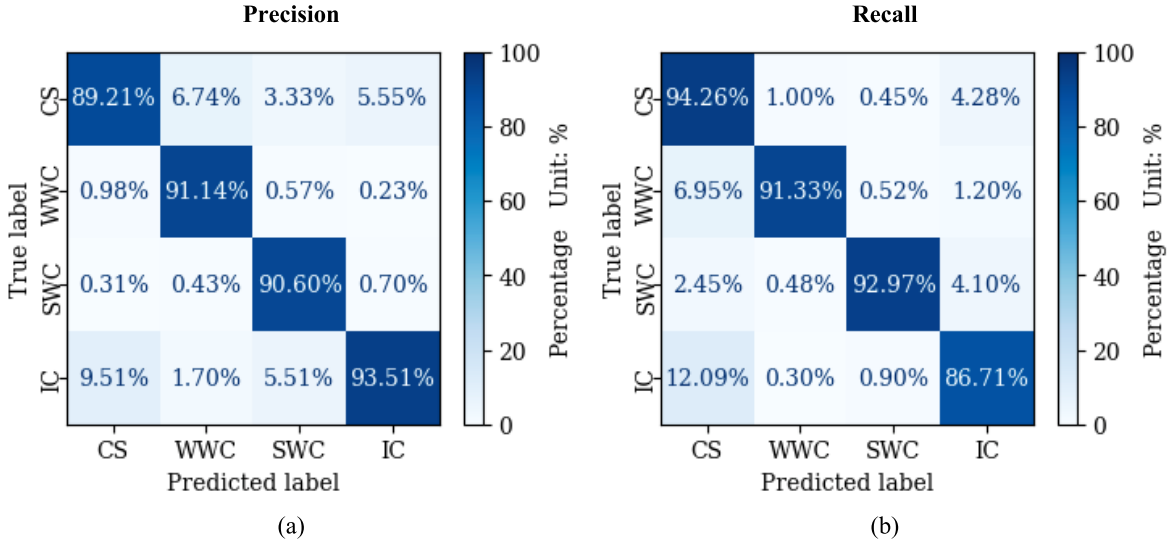


Fig. 2. Flowchart of the SWC detection algorithm using AHI, MODIS, and ERA5 data.



(a) (b)

Fig. 3. Confusion matrix of the model using an independent dataset from January, April, August, and November 2017. (a) Precisions. (b) Recalls on the diagonal.

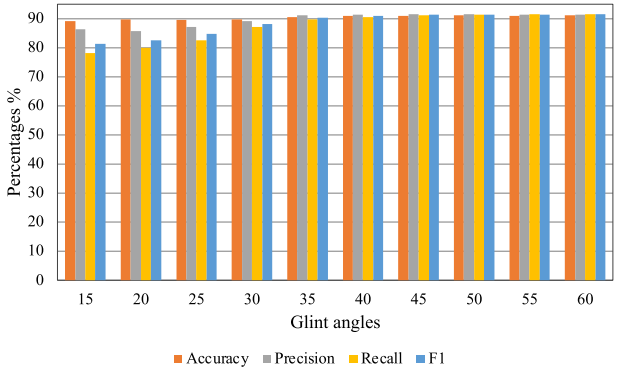


Fig. 4. Model performance metrics at various sun glint angle thresholds.

model maintains high overall accuracy, indicating its robust performance.

TABLE II
MODEL EVALUATION ACROSS VARIOUS CONDITIONS

Model	Accuracy	Precision	Recall	F1
All	91.00%	90.60%	92.97%	91.77%
Land	**91.84%**	89.63%	89.35%	89.49%
Water	90.75%	**90.81%**	**93.78%**	**92.27%**
Day	**91.95%**	**90.90%**	92.03%	91.46%
Night	89.91%	90.23%	**94.14%**	**92.14%**

The model's performance across different conditions is analyzed in Table II, revealing only minor variations in surface and diurnal metrics that are not significantly different from the overall results, as illustrated by the confusion matrices in Figs. S3 and S4 in the Supplementary Material. Additionally, no significant seasonal variations were observed,

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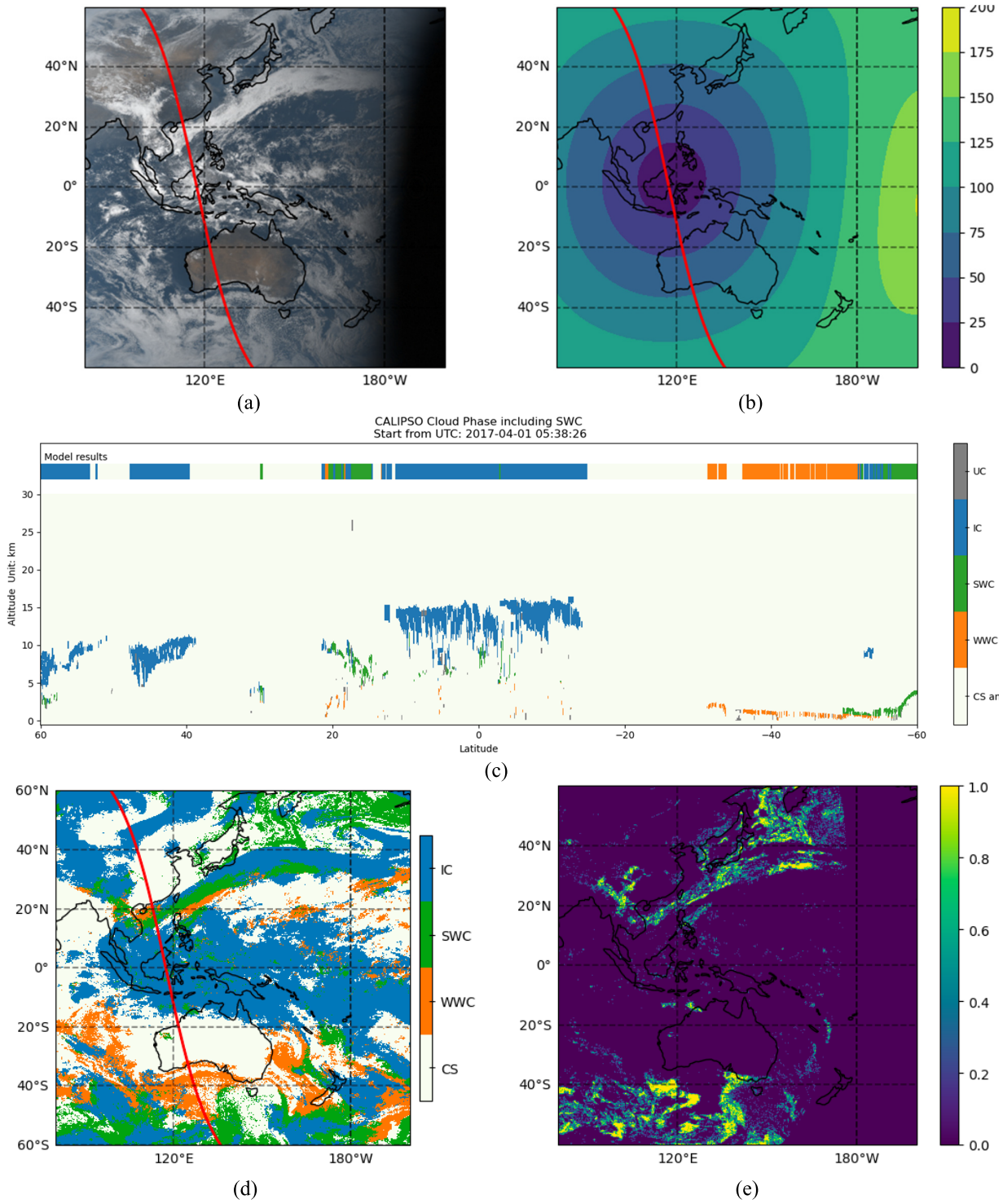


Fig. 5. Case study in sun glint region. (a) True color composition of Himawari-8/AHI imagery. (b) Sun glint angle calculations. (c) Comparison of CALIPSO CPH profiles with model outcomes, where light green represents CS or no data, orange for WWCs, green for SWCs, blue for ICs, and gray for unclassified. (d) Results by the proposed model. (e) SWC detection results using the algorithm from Wang et al. [29].

with these findings detailed in Fig. S5 in the Supplementary Material. The confusion matrix of the model trained without SMOTE is shown in Fig. S6. Its accuracy is high and even achieves a higher precision. However, the recalls decreased significantly, especially for WWC and SWC, indicating that the results are affected by the imbalance of the training set.

Sun glint significantly impacts the VNIR bands, affecting the performance of cloud retrieval algorithms. Table III details how different glint angle thresholds influence the failure rates of Himawari-8/AHI cloud products in detecting SWCs in the testing dataset. For instance, at a 40-° glint angle, 21.54% of the 124 685 cloud samples within the sun glint region fail detection, constituting 95.17% of potential SWCs samples.

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TABLE III
RESULTS OF SWC DETECTION USING HIMAWARI-8 CLOUD PRODUCTS IN THE SUN GLINT REGION

Glint Angle Threshold (°)	Total Clouds	Invalid Products in Total Clouds (%)	Invalid Products in Potential SWCs (%)
15	17877	26.14	96.77
20	31787	26.41	96.52
25	48824	26.47	96.70
30	70299	25.78	96.55
35	95268	23.71	96.05
40	124685	21.54	95.17
45	157546	19.74	93.52
50	190353	18.43	91.51
55	223361	17.80	89.48
60	258301	17.39	86.20

This issue is exacerbated as the glint angle narrows, reducing the algorithm's SWCs identification accuracy. However, the algorithm proposed in this study, utilizing the TIR channel to detect blackbody radiation, remains unaffected by sun glint. Further evaluations under various glint angles in Fig. 4 show a slight decline in evaluation metrics, yet the model consistently performs well without failure.

B. Outcomes From a Full Disk Case Study

Fig. 5 showcases a case study of sun glint impact, comparing model predictions with CALIPSO observations and analyzing spatial distribution with others. The case, observed at 5:38 UTC on April 1, 2017, includes a true color composite from Himawari-8/AHI [Fig. 5(a)], with CALIPSO's track highlighted, and a sun glint angle calculation [Fig. 5(b)], illustrating a significant glint at the observation time. As shown in Fig. 5(c), the analysis demonstrates that the model aligns closely with CALIPSO's CPH data, accurately identifying cloud types including a notable detection of SWCs near 30°N. Despite challenging conditions between 20°N and 10°N, including several cloud types mixed, the model maintains high accuracy. At 0°N, within the intense sun glint region, the model's results still match CALIPSO observations, underscoring its robustness and effectiveness even under strong interference. Fig. 5(d) and (e) present full disk results of SWCs, respectively, showing consistency with Wang's method in daytime regions and demonstrating the model's capability at night. Unshown cases also yielded similarly positive outcomes, aligning with CALIPSO data across different conditions.

VI. CONCLUSION

SWCs play a crucial role in the Earth's radiation balance, aviation safety, and the effectiveness of artificial precipitation techniques. This study utilized TIR channels from AHI onboard the geostationary satellite Himawari-8 for identifying SWCs, with CALIPSO/CALIOP observations serving as model and validation benchmarks. By analyzing the complex refraction indices of ice and supercooled water in the TIR spectrum and employing PCA, the sensitivity in distinguishing these two types was determined. The machine learning model incorporated radiative transfer-related input variables, all with well-defined physical interpretations, to ensure strong generalization across different scenarios. Evaluated with a time-independent dataset, the model accurately identified SWCs in a variety of conditions, effectively overcoming limitations of visible light algorithms at night and under sun glint interference.

The model's effectiveness is yet to be fully assessed in environments with significant disruptions, like dense aerosol pollution or extensive ice/snow coverage. Its reliance on single-point data restricts its capacity to leverage the continuous temporal and spatial data available from high-frequency geostationary satellite observations, which could limit its accuracy in tracking dynamic cloud movements and transformations. Future research aims to explore the potential of integrating temporal and spatial data to enhance the precision, speed, and convenience of SWC detection.

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